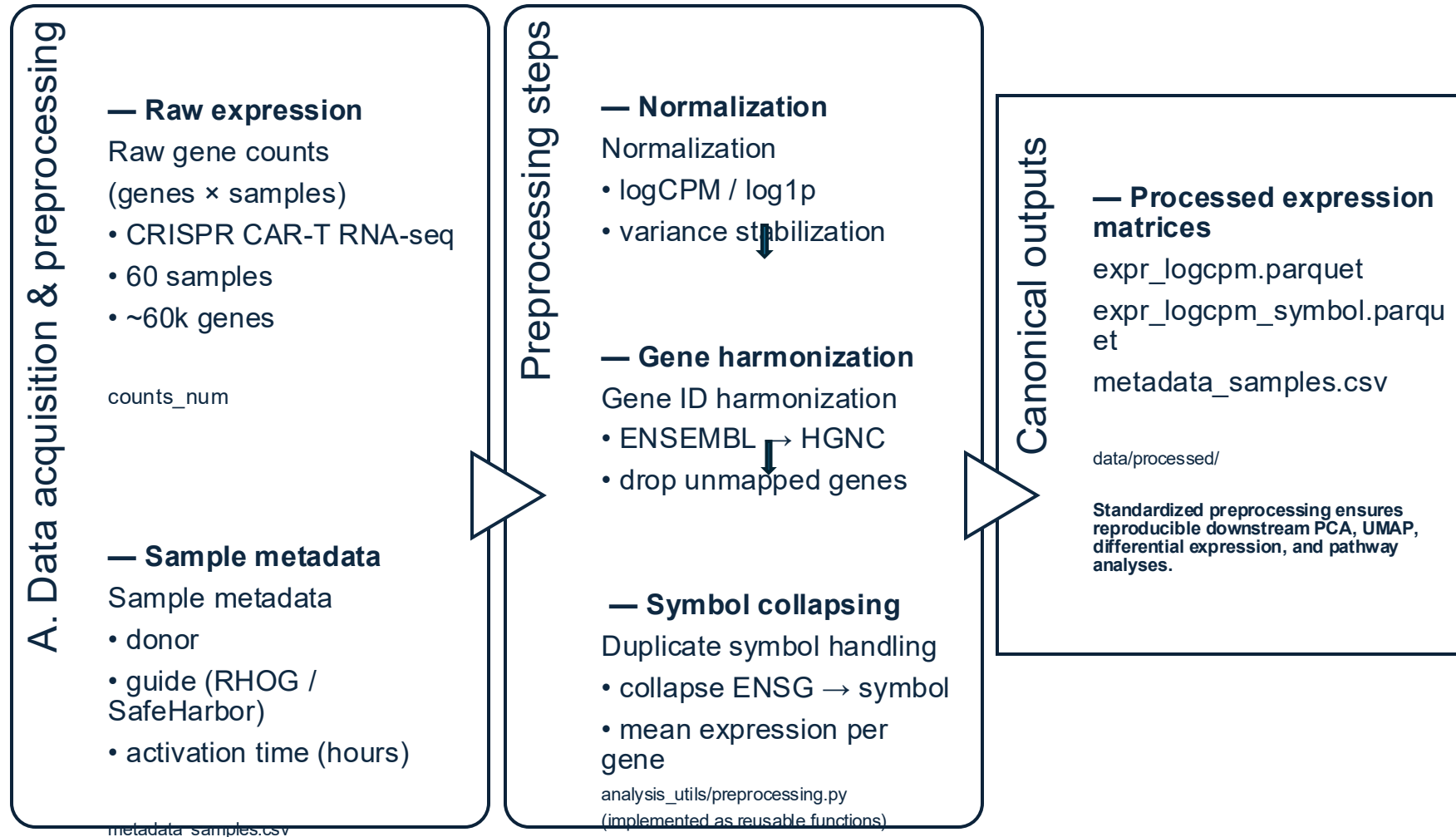


- **Figure 1 = Panels A–C**

Panel	Content	Source
A	Data acquisition & preprocessing schematic	Manual (PowerPoint)
B	PCA + UMAP workflow + biological axis interpretation	Manual (PowerPoint)
C	Differential expression → pathway abstraction	Manual (PowerPoint)

A. Data acquisition & preprocessing Schematic



B. Low-dimensional representations for trajectory and structure analysis

PCA computation-outputs

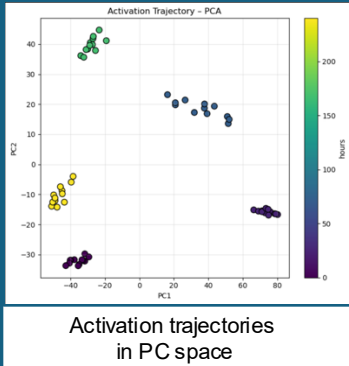
PCA Biological Interpretation of PCA axis

Input

Symbol-level normalized expression

Principal Component Analysis (PCA) • samples × genes • variance decomposition <code>analysis_utils/pca_utils.py</code>	
PC scores (samples x PCs) <code>pca_scores.parquet</code> <code>pca_loadings.parquet</code>	PC loadings (genes × PCs) <code>pca_scores.parquet</code> <code>pca_loadings.parquet</code>

- **PC1:** →
Effector IL2/STAT5 & NFκB activation
↔ *Interferon-driven activation*
- **PC2:** ↑
Interferon stress
↔ *Proliferative E2F / OxPhos programs*



Dimensionality reduction outputs

- PCA figures (results/pca/)
- UMAP figures (results/umap/)
- Metadata with PCs (meta_with_pcs)

Reusable for DE, pathway scoring, and trajectory analysis

- logCPM expression
- genes × samples

UMAP

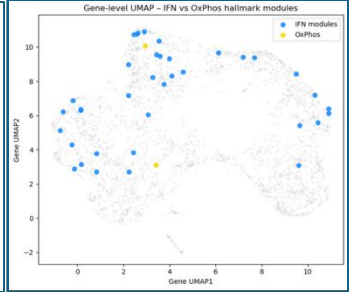
`expr_logcpm_symbol.parquet`

Feature selection • highly variable genes (HVGs)	UMAP embedding UMAP (sample-level) • nonlinear manifold learning • preserves local structure <code>analysis_utils/umap_utils.py</code>
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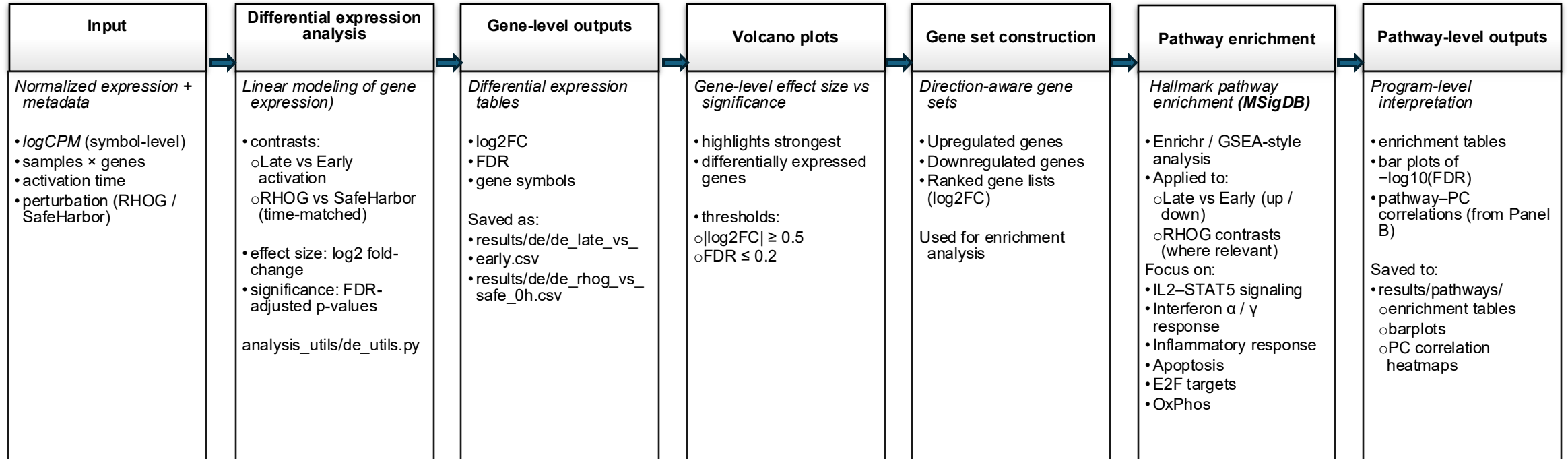
UMAP outputs

UMAP coordinates (samples × 2)

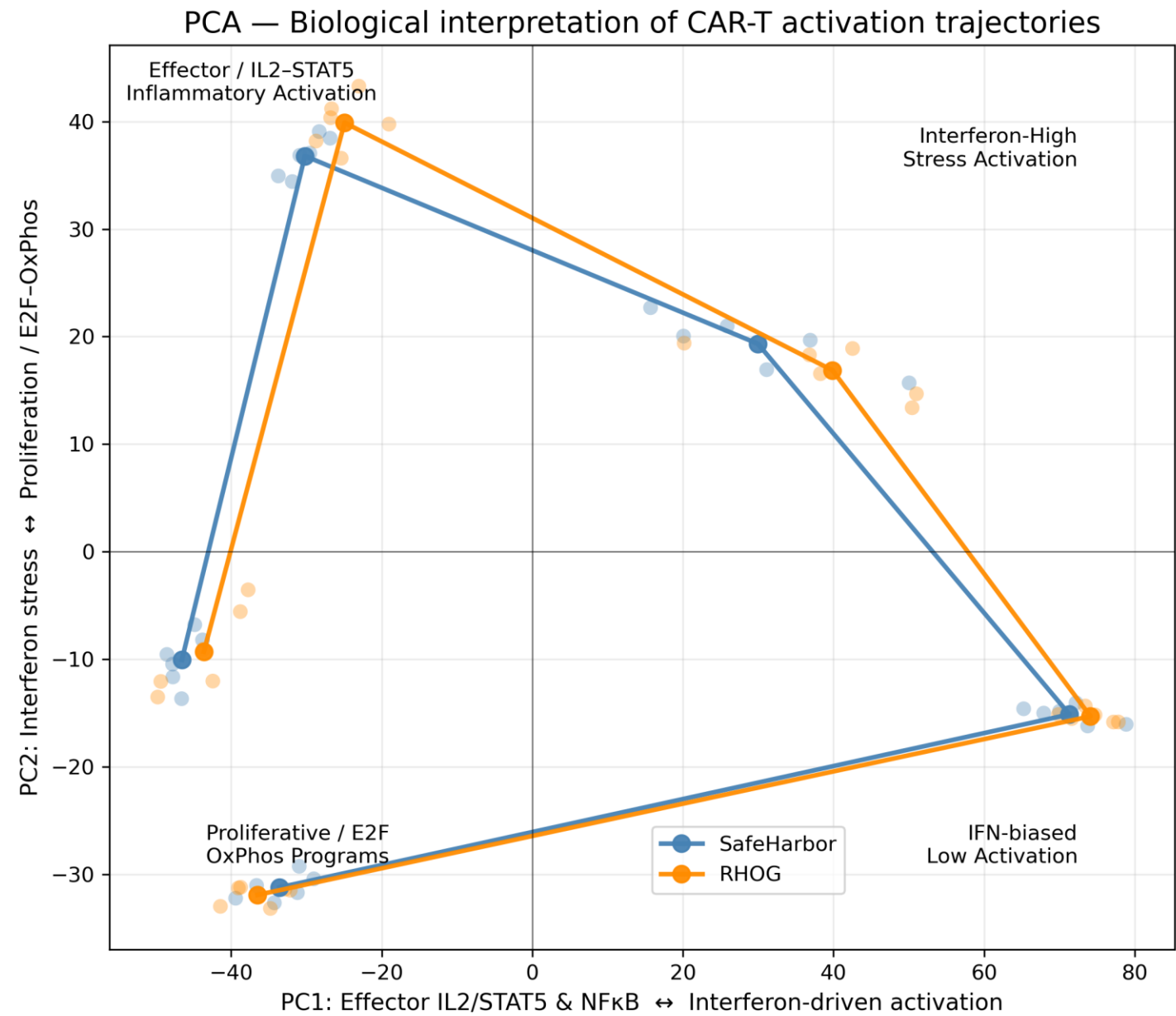
`sample_umap_coordinates.csv`



Panel C – Differential expression → pathway abstraction



Panel D – Pathways & Module Scores



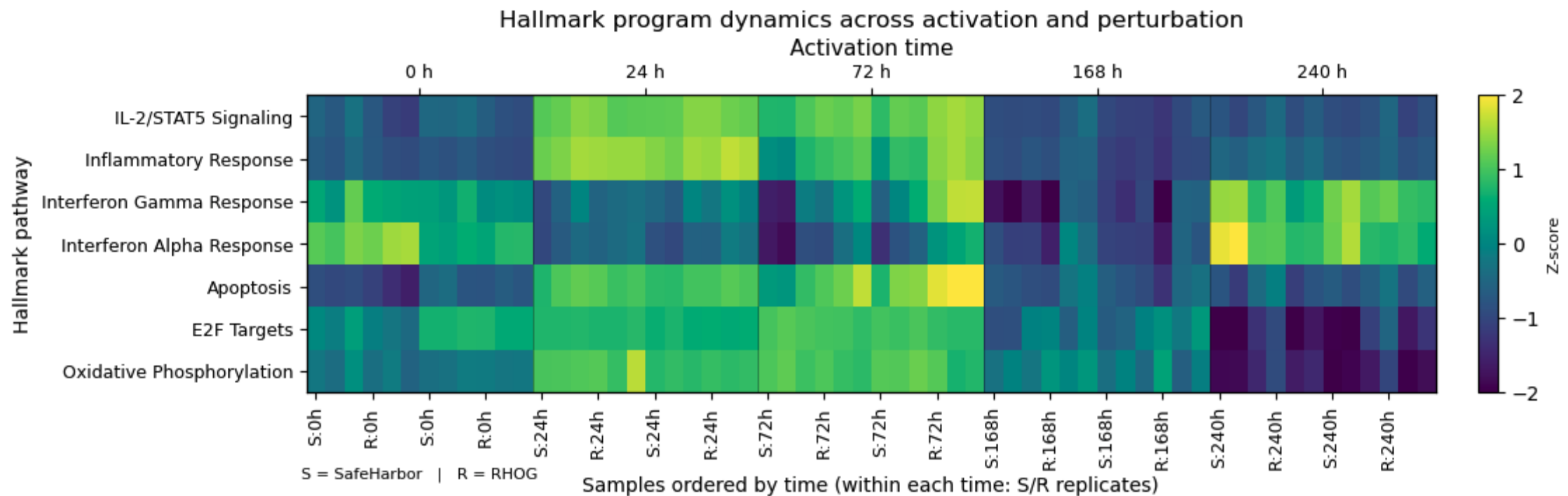


Figure A05. Hallmark pathway dynamics across CAR-T cell activation and RHOG perturbation.

Heatmap showing z-scored Hallmark pathway module scores computed from symbol-level logCPM expression across activation time (0–240 h). Samples are ordered by time, with SafeHarbor (S) and RHOG-knockout (R) biological replicates shown within each time point (n = 6 donors per condition per time). Across donors, CAR-T activation induces coordinated IL-2/STAT5, inflammatory, interferon, apoptotic, E2F, and oxidative phosphorylation programs, revealing a highly conserved temporal structure. RHOG perturbation subtly but consistently reshapes these trajectories, most prominently at late activation stages, highlighting pathway-level effects that are robust to donor-to-donor variability.